SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS
FINAL ASSESSMENT TEST
WINTER SEMESTER 2025-2026

Programme Name & Branch	**: M.Tech (SE)**
Course Code and Course Name	**: ISWE406P – AGILE DEVELOPMENT PROCESS & DEVOPS LAB**
Faculty Name(s)	**: Prof. Sweta Bhattacharya**
Class Number(s)	**: 2390**
Slot	**: L3+L4**
Date of Examination	**: 13.04.26**

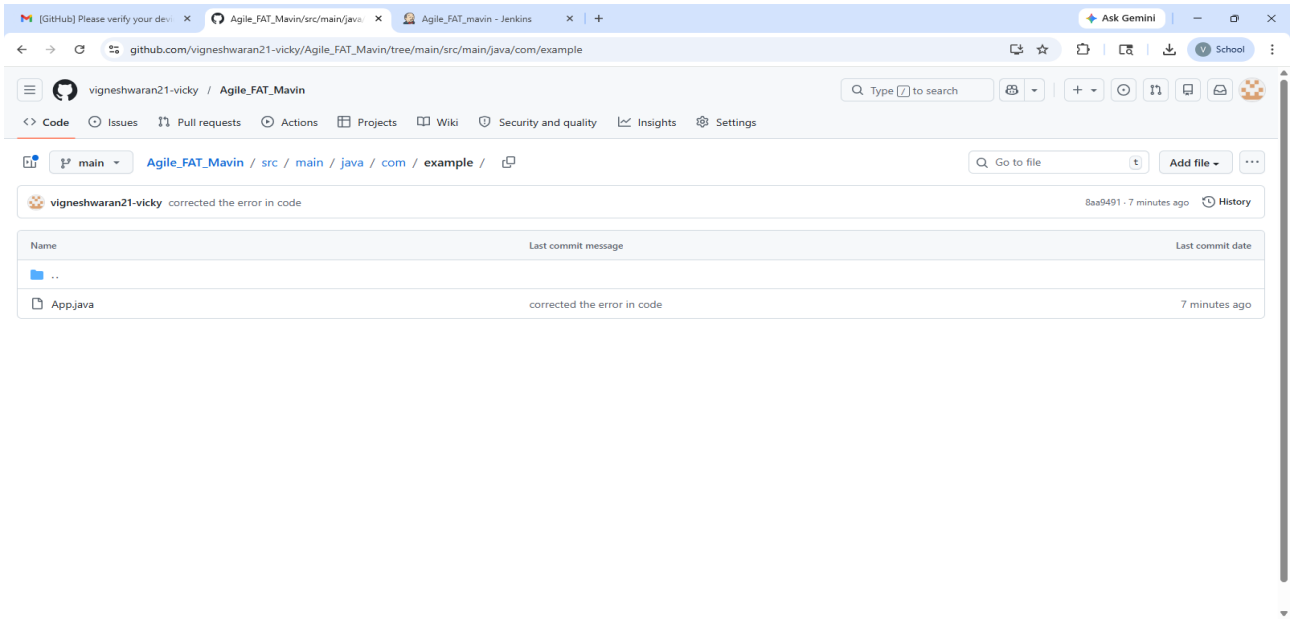
Name: VIGNESHWARAN T
Reg no: 23MIS0587

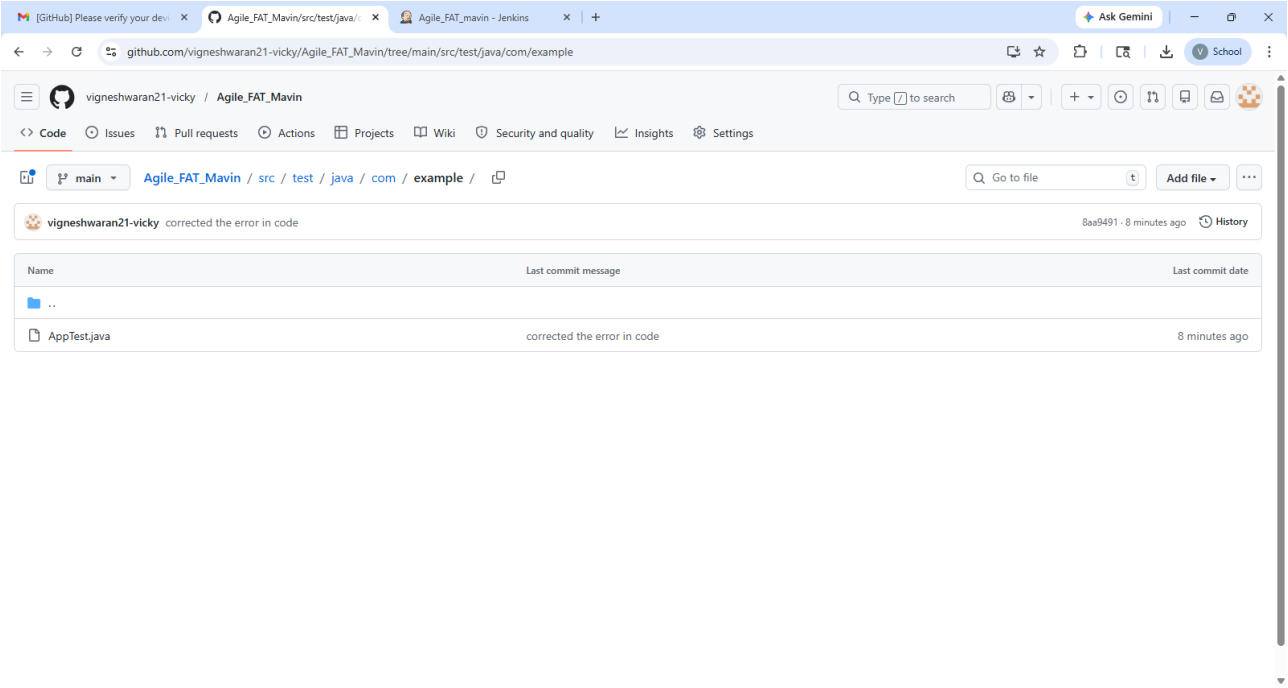
QUESTION:

Design and implement a Voting Eligibility System program that determines whether a person is eligible to vote based on specific criteria. The program should define an appropriate class to store voter details such as name, age, citizenship, voter ID, and ID validity status. It must accept input for one or more voters and evaluate eligibility using the conditions that the voter must be at least 18 years old, must be a citizen of the country (e.g., Indian citizen), and must possess a valid voter ID. The program should display whether each voter is eligible or not, and in cases of ineligibility, it should clearly indicate the reason, such as being underage, not a citizen, or having an invalid ID

1. Create a GitHub repository for a Maven-based application following the standard Maven directory structure [10]

GitHub repository link: https://github.com/vigneshwaran21-vicky/Agile_FAT_Mavin.git

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Agile_FAT_mavin - Jenkins
Ask Gemini
github.com/vigneshwaran21-vicky/Agile_FAT_Mavin/tree/main/src/test/java/com/example
School
vigneshwaran21-vicky / Agile_FAT_Mavin
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Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
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8aa9491 · 8 minutes ago
History
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Last commit message
Last commit date
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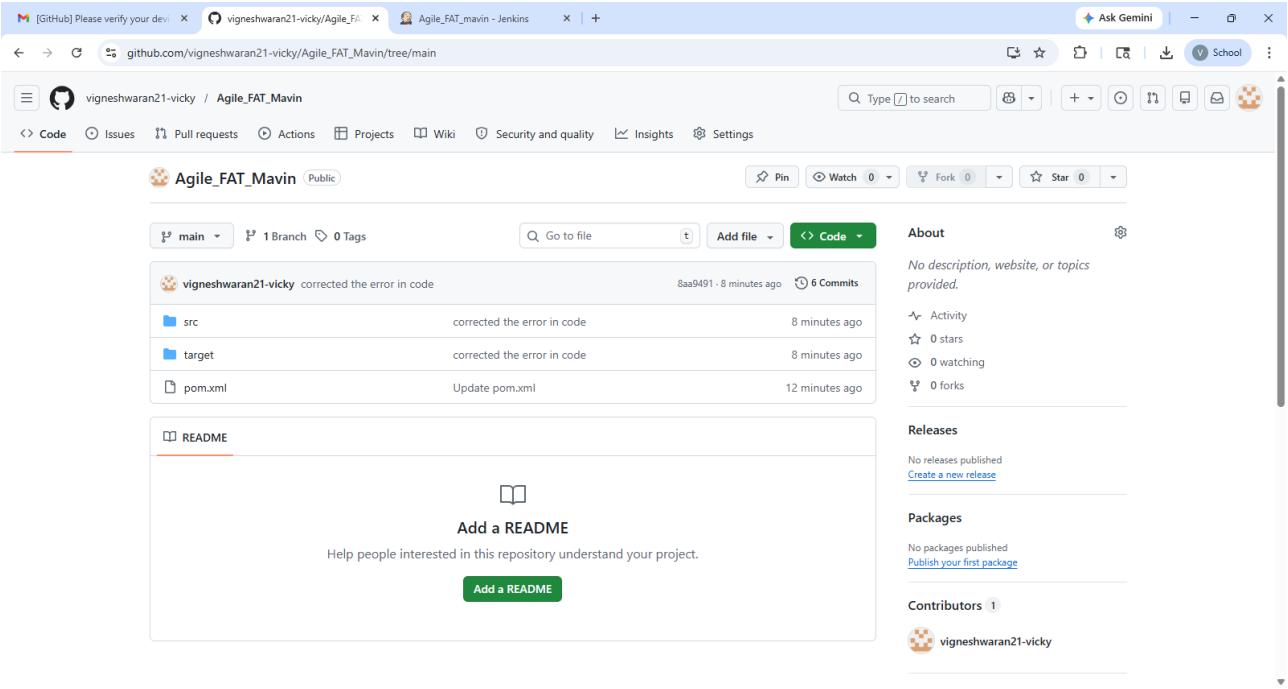
AppTest.java
corrected the error in code
8 minutes ago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Agile_FAT_mavin - Jenkins
Ask Gemini
github.com/vigneshwaran21-vicky/Agile_FAT_Mavin/tree/main
School
vigneshwaran21-vicky / Agile_FAT_Mavin
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
Agile_FAT_Mavin
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
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Code
vigneshwaran21-vicky corrected the error in code
8aa9491 · 8 minutes ago
6 Commits
src
corrected the error in code
8 minutes ago
target
corrected the error in code
8 minutes ago
pom.xml
Update pom.xml
12 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
vigneshwaran21-vicky

Codes:-

src/main/java/com/example/App.java

```
package com.example;
import java.util.Scanner;
public class App {
    public static String getEligibilityStatus(int age, boolean isCitizen, boolean hasValidID) {
        if (age < 18) return "Ineligible (Reason: Underage)";
        if (!isCitizen) return "Ineligible (Reason: Not a citizen)";
        if (!hasValidID) return "Ineligible (Reason: Invalid ID)";
        return "Eligible";
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("--- Voting Eligibility System ---");
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Age: ");
        int age = sc.nextInt();
        System.out.print("Is Citizen (true/false): ");
        boolean citizen = sc.nextBoolean();
        System.out.print("Is ID Valid (true/false): ");
        boolean validID = sc.nextBoolean();
        String status = getEligibilityStatus(age, citizen, validID);
        System.out.println("\nVoter: " + name);
        System.out.println("Status: " + status);
        sc.close();
    }
}
```

src/test/java/com/example/AppTest.java

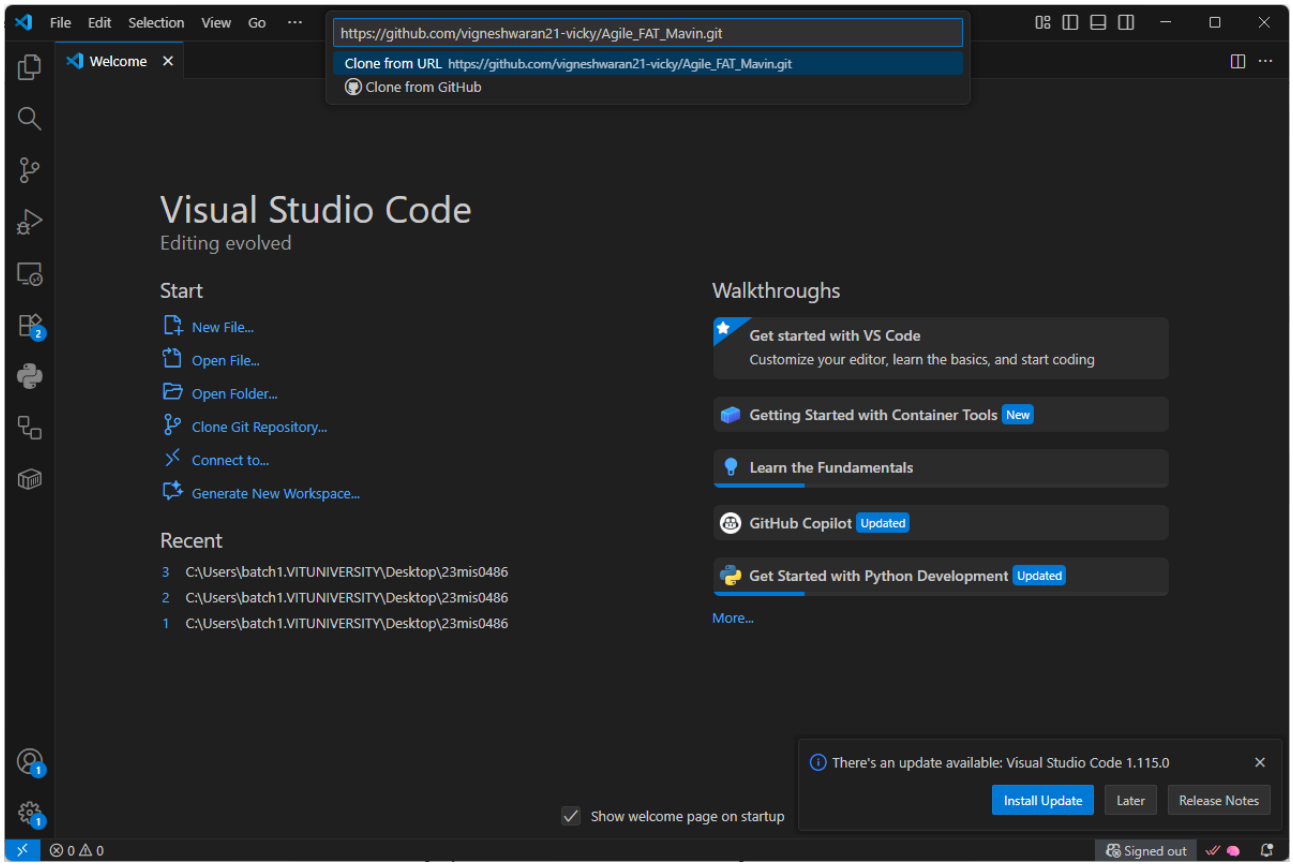
```
package com.example;
import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;
public class AppTest {
    @Test
    public void testEligibleVoter() {
        assertEquals("Eligible", App.getEligibilityStatus(25, true, true));
    }
    @Test
    public void testUnderageVoter() {
        assertEquals("Ineligible (Reason: Underage)", App.getEligibilityStatus(17, true, true));
    }
    @Test
    public void testNonCitizen() {
        assertEquals("Ineligible (Reason: Not a citizen)", App.getEligibilityStatus(30, false, true));
    }
    @Test
    public void testInvalidID() {
        assertEquals("Ineligible (Reason: Invalid ID)", App.getEligibilityStatus(20, true, false));
```

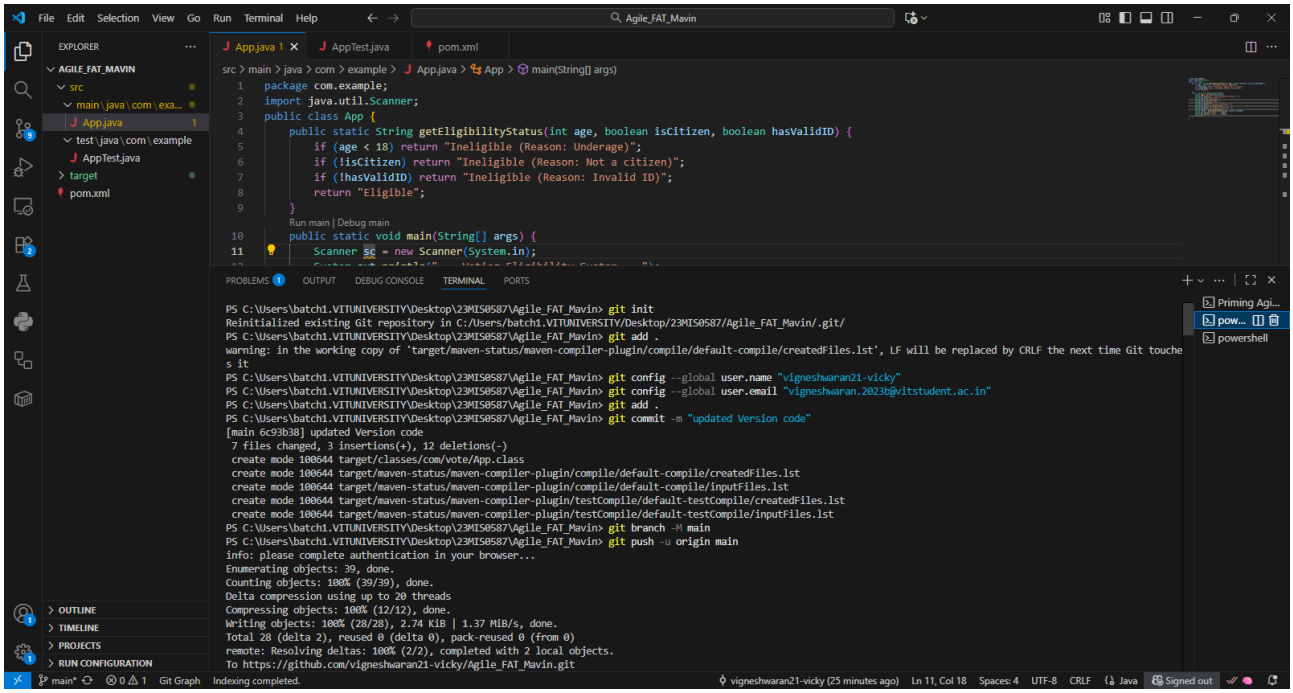
```
    }
}
```

pom.xml

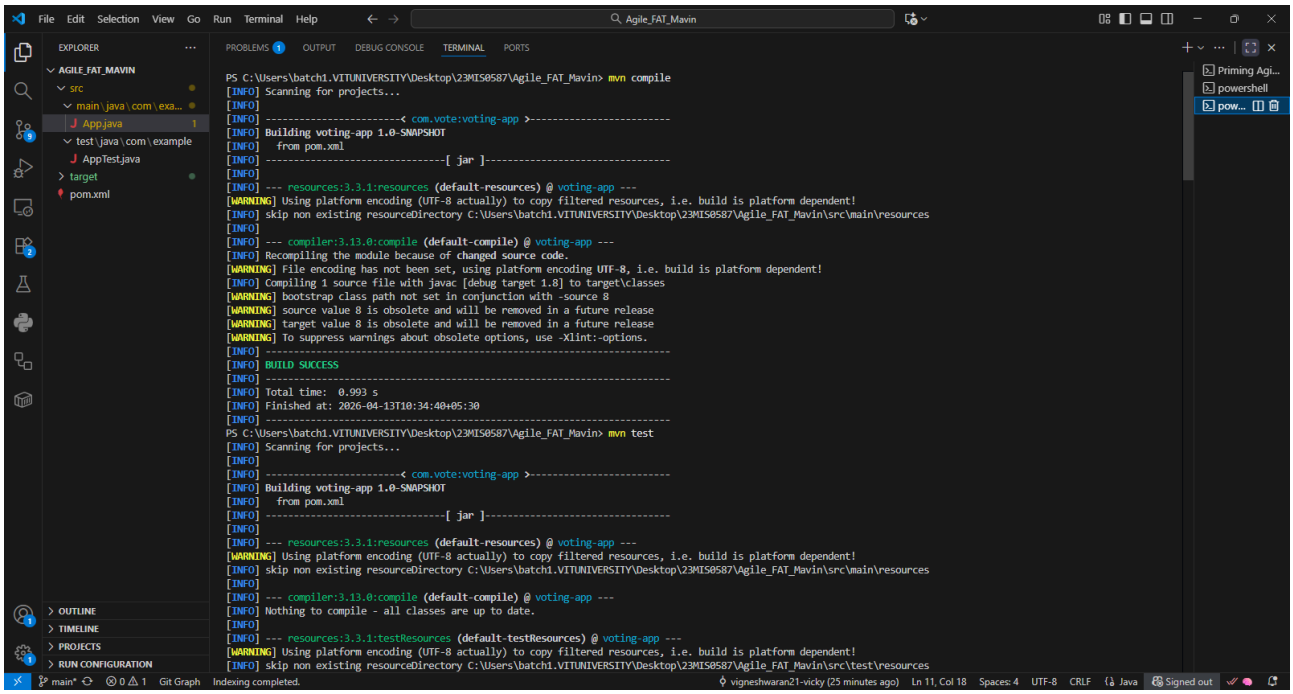
```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.vote</groupId>
    <artifactId>voting-app</artifactId>
    <version>1.0-SNAPSHOT</version>


    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.10.0</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.0.0</version>
            </plugin>
        </plugins>
    </build>

</project>
```

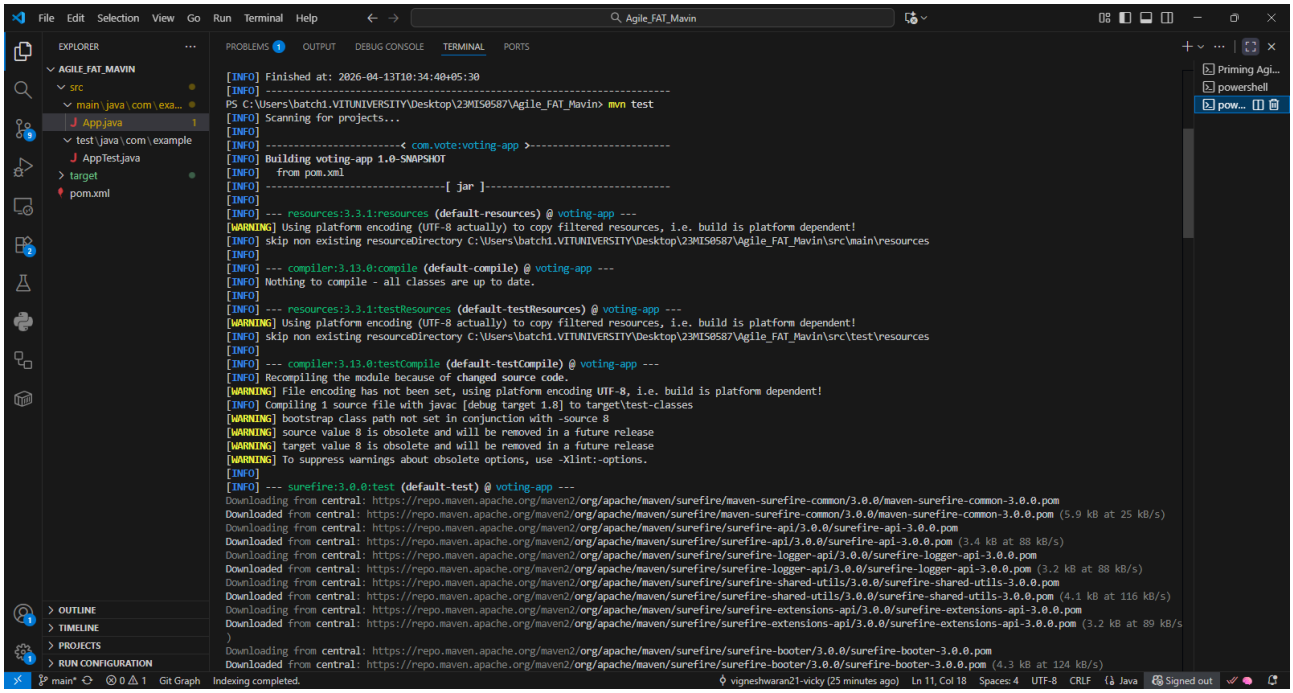
2. Clone this repository into VS Code, make the necessary code modifications, and commit the changes back to the Git repository [10]





3. Build and verify the application within VS Code to ensure it is compatible with Jenkins integration [10]

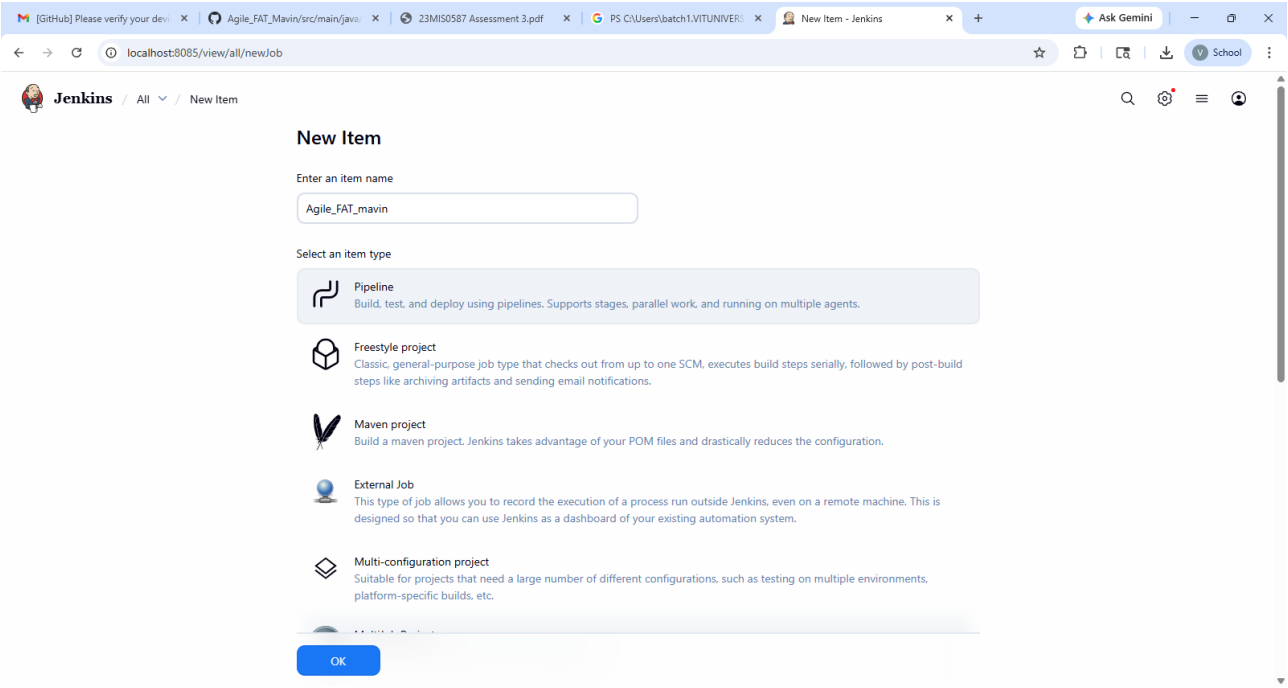


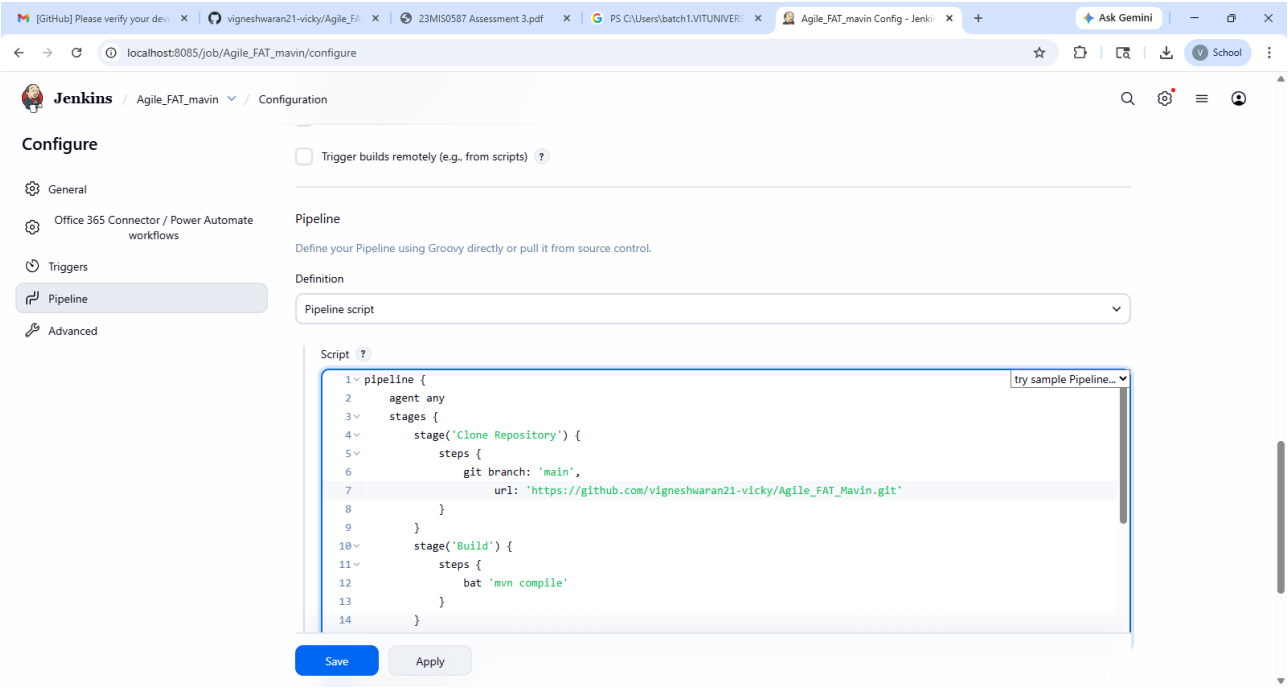


```
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.044 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.733 s
[INFO] Finished at: 2026-04-13T10:34:53+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin> mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.vote:voting-app >------------------------
[INFO] Building voting-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ voting-app ---
[INFO] Deleting C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ voting-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ voting-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ voting-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ voting-app ---
```

```
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.0.0:test (default-test) @ voting-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.047 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ voting-app ---
[INFO] Building jar: C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin\target\voting-app-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ voting-app ---
[INFO] Installing C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin\pom.xml to C:\Users\batch1.VITUNIVERSITY\.m2\repository\com\vote\voting-app\1.0-SNAPSHOT
\voting-app-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin\target\voting-app-1.0-SNAPSHOT.jar to C:\Users\batch1.VITUNIVERSITY\.m2\repository\com\v
ote\voting-app\1.0-SNAPSHOT\voting-app-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.540 s
[INFO] Finished at: 2026-04-13T10:35:03+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\batch1.VITUNIVERSITY\Desktop\23MIS0587\Agile_FAT_Mavin>
```

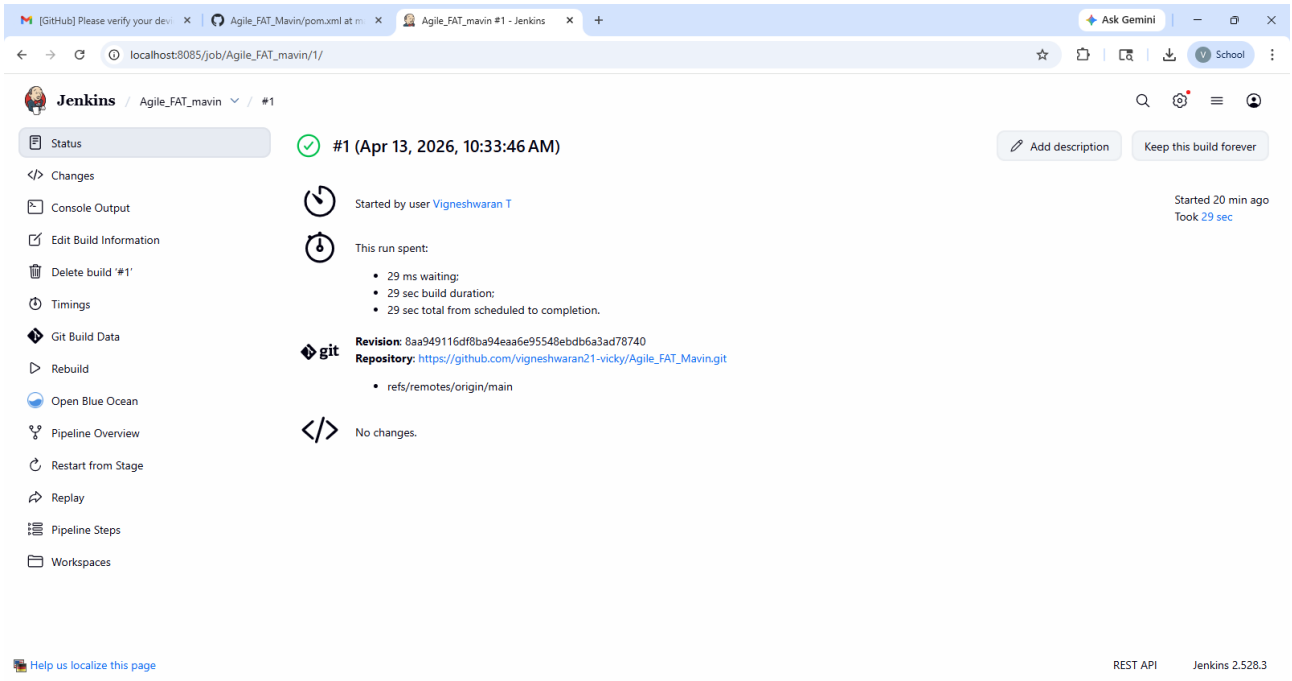
4. Finally, set up a Pipeline project in Jenkins and execute the CI/CD process by writing and running the required pipeline script [20]

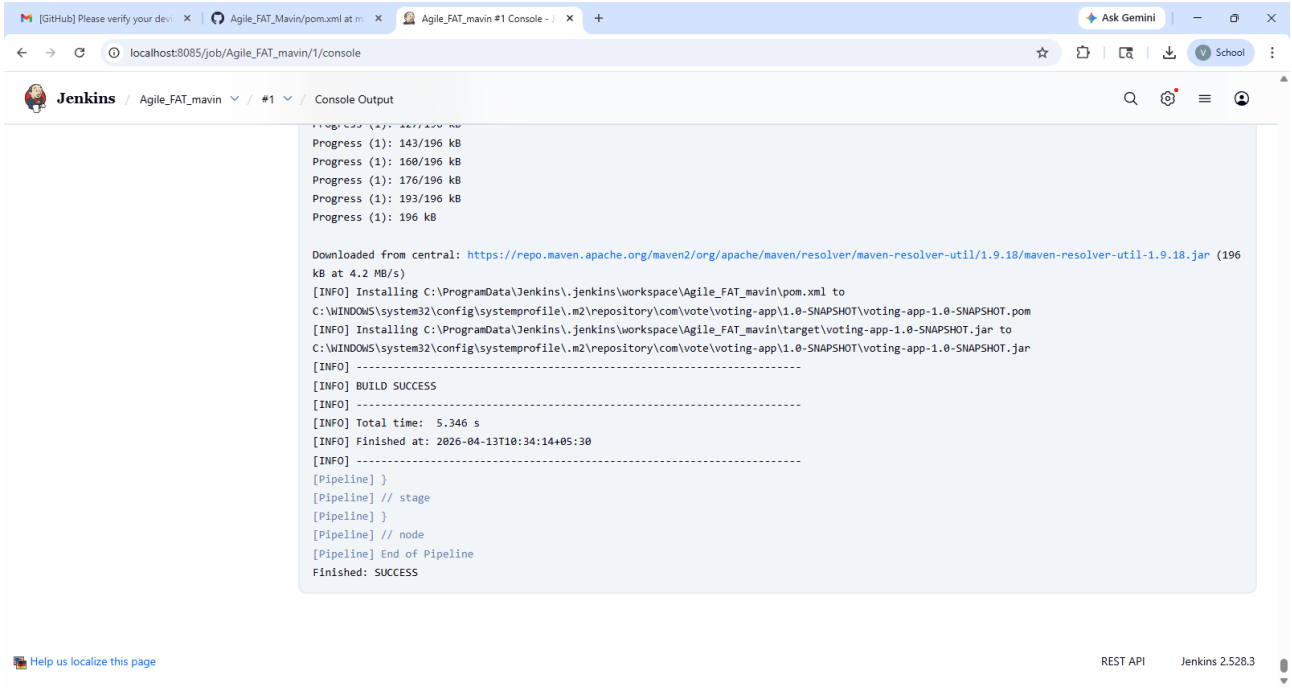


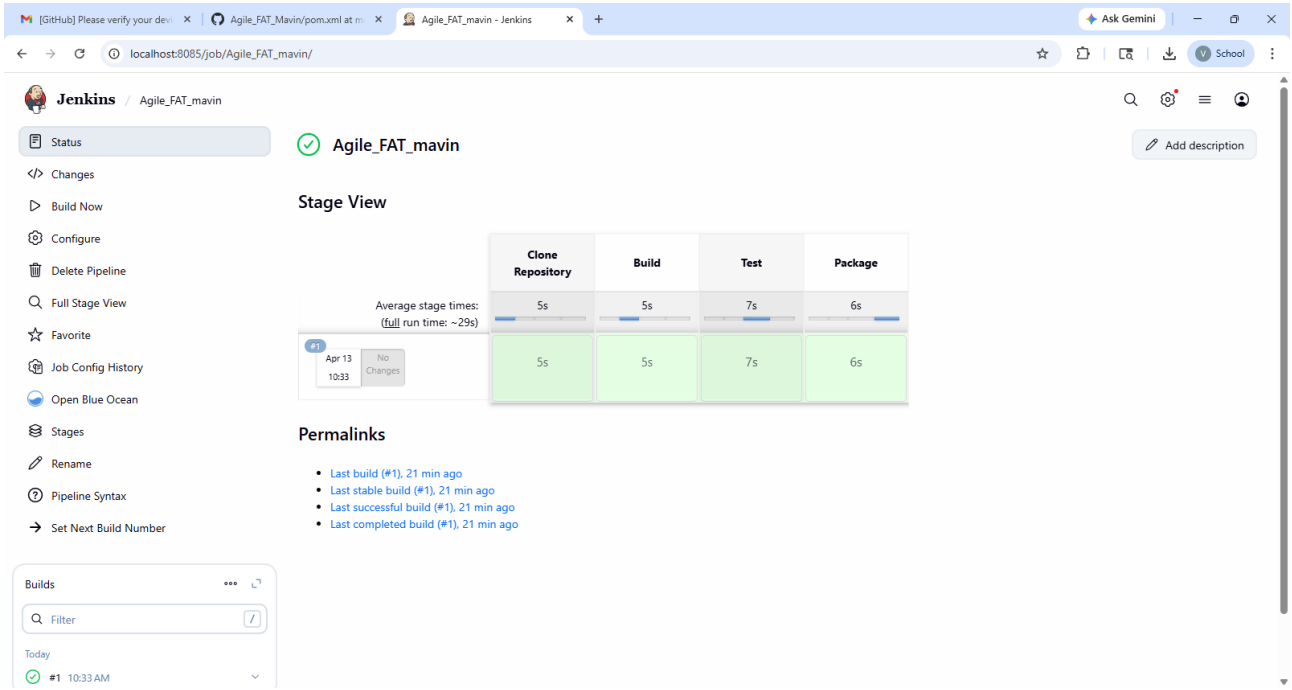


Pipeline Script:-

```
pipeline {
   agent any
   stages {
      stage('Clone Repository') {
         steps {
            git branch: 'main',
                 url: 'https://github.com/vigneshwaran21-vicky/Agile_FAT_Mavin.git'
         }
      }
      stage('Build') {
         steps {
            bat 'mvn compile'
         }
      }

      stage('Test') {
         steps {
            bat 'mvn test'
         }
      }
      stage('Package') {
         steps {
            bat 'mvn clean install'
         }
      }
   }
}
```




[GitHub] Please verify your devi
Agile_FAT_Mavin/pom.xml at m
Agile_FAT_mavin #1 Console - J
Ask Gemini
localhost:8085/job/Agile_FAT_mavin/1/console
School
Jenkins / Agile_FAT_mavin / #1 / Console Output
Progress (1): 143/196 kB
Progress (1): 160/196 kB
Progress (1): 176/196 kB
Progress (1): 193/196 kB
Progress (1): 196 kB
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 4.2 MB/s)
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Agile_FAT_mavin\pom.xml to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\vote\voting-app\1.0-SNAPSHOT\voting-app-1.0-SNAPSHOT.pom
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Agile_FAT_mavin\target\voting-app-1.0-SNAPSHOT.jar to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\vote\voting-app\1.0-SNAPSHOT\voting-app-1.0-SNAPSHOT.jar
[INFO] --
[INFO] BUILD SUCCESS
[INFO] --
[INFO] Total time: 5.346 s
[INFO] Finished at: 2026-04-13T10:34:14+05:30
[INFO] --
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Help us localize this page
REST API
Jenkins 2.528.3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Agile_FAT_mavin - Jenkins
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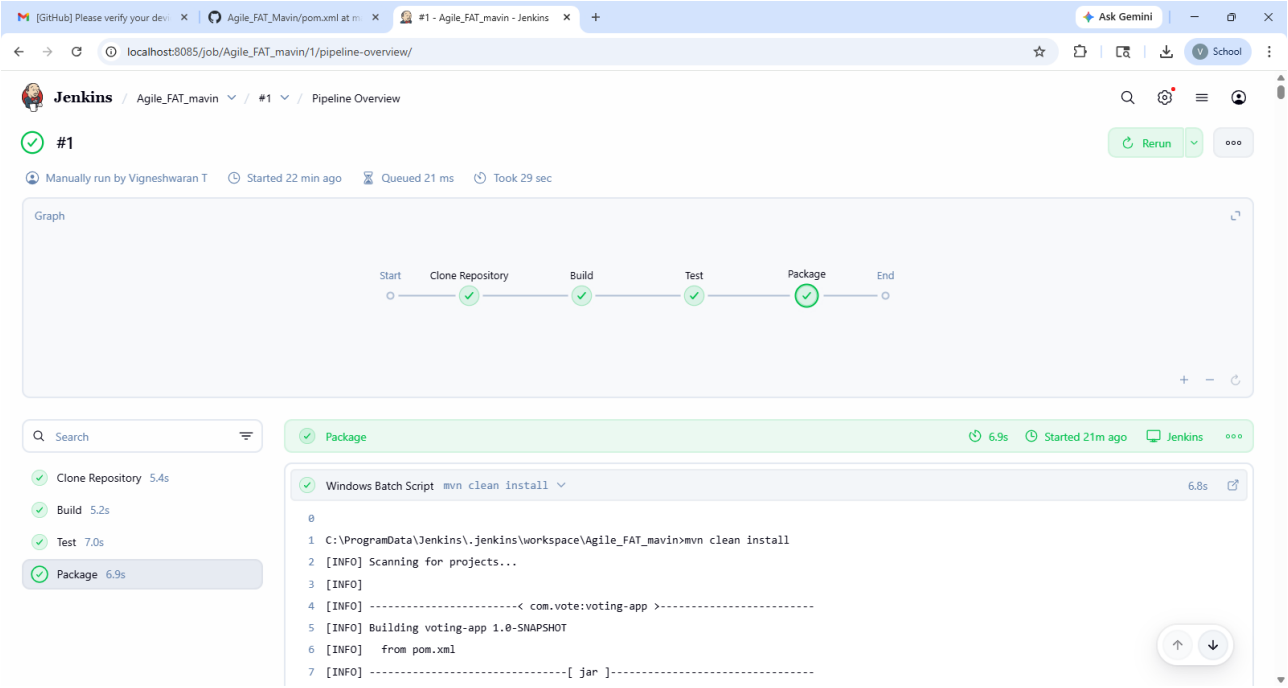
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Favorite
Job Config History
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Set Next Build Number
Builds
Filter
Today
#1 10:33 AM
Agile_FAT_mavin
Add description
Stage View
Clone Repository
Build
Test
Package
Average stage times:
(full run time: ~29s)
5s
5s
7s
6s
#1
Apr 13
10:33
No Changes
5s
5s
7s
6s
Permalinks
Last build (#1), 21 min ago
Last stable build (#1), 21 min ago
Last successful build (#1), 21 min ago
Last completed build (#1), 21 min ago


[GitHub] Please verify your devi
Agile_FAT_Mavin/pom.xml at m
#1 - Agile_FAT_mavin - Jenkins
Ask Gemini
localhost:8085/job/Agile_FAT_mavin/1/pipeline-overview/
School
Jenkins / Agile_FAT_mavin / #1 / Pipeline Overview
#1
Rerun
Manually run by Vigneshwaran T
Started 22 min ago
Queued 21 ms
Took 29 sec
Graph
Start
Clone Repository
Build
Test
Package
End
Search
Clone Repository 5.4s
Build 5.2s
Test 7.0s
Package 6.9s
Package
6.9s
Started 21m ago
Jenkins
Windows Batch Script mvn clean install
6.8s
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\Agile_FAT_mavin>mvn clean install
2 [INFO] Scanning for projects...
3 [INFO]
4 [INFO] ------------------------< com.vote:voting-app >-------------------------
5 [INFO] Building voting-app 1.0-SNAPSHOT
6 [INFO] from pom.xml
7 [INFO] --------------------------------[jar]---------------------------------